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Lanostane-type C_{31} triterpenoid derivatives from the fruiting bodies of cultivated *Fomitopsis palustris*

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ABSTRACT

Fifteen undescribed and five known lanostane-type C_{31} triterpenoid derivatives were isolated from the aqueous EtOH extract of the fruiting bodies of cultivated *Fomitopsis palustris*. Their structures were identified from the spectroscopic data and chemical degradation studies. The structures of palustrisoic acids A and H were confirmed by X-ray crystallography. Polyporenic acid B showed strong cytotoxicity against the HCT116, A549, and HepG2 cell lines with IC_{50} values of 8.4, 12.1, and 12.2 μM, respectively. Palustrisolides A, C, and G displayed weak cytotoxicity.

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1. Introduction

The species of *Fomitopsis* (family Fomitopsidaceae) have been used for centuries as popular medicines to prevent and/or treat different diseases in Asia. Chemical studies on the fungi of this genus led to the identification of several classes of natural products such as chlorinated coumarins, steroids, lanostane-type triterpenoids, polyphenols, and polyketides (Chiba et al., 2014; De Silva et al., 2013; Hwang et al., 2013; Keller et al., 1996; Pleszczyńska et al., 2017; Quang et al., 2005; Rösecke and König, 1999; Shi et al., 2017; Yoshikawa et al., 2005). Some of them showed cholesterol reduction (Chiba et al., 2014), antiviral and antibacterial (Keller et al., 1996; Pleszczyńska et al., 2017), antitumor (Shi et al., 2017), and anti-inflammatory activities (Yoshikawa et al., 2005; Pleszczyńska et al., 2017). *F. palustris* was reported to cause wood brown rot, a disease resulting from the enzymatic breakdown of cellulose, the major wood component (Konuma et al., 2015). The extracts of the fruiting bodies of cultivated *F. palustris* showed cytotoxicity against different cancer cell lines (Yu et al., 2014). The preliminary results revealed that the aforementioned extracts inhibited the growth of tumor cells and promoted the apoptosis of tumor cells (Yu et al., 2014). In the course of the chemical investigations on the cytotoxic extract of the fruiting bodies of cultivated fungus *F. palustris*, 20 lanostane-type C_{31} triterpenoid derivatives were obtained and identified from the spectroscopic data and alkaline hydrolysis experiments. Lanostane-type triterpenoids are well-known tetracyclic triterpene derivatives (Hill and Connolly, 2017; Nes, 2011). Most of them showed very interesting biological activities, such as anti-AIDS (Li et al., 1993), anti-inflammatory (Kamo et al., 2003), and cytotoxicity (Lai et al., 2016; Tohtahon et al., 2017). Herein we report the isolation, structure elucidation, and cytotoxicity of these lanostane-type C_{31} triterpenoid derivatives.

2. Results and discussion

The 95% EtOH extract of the fruiting bodies of cultivated *F. palustris* was partitioned between H_2O and EtOAc to afford the

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crude cytotoxic residue. The residue was separated and purified by column chromatography and semi-preparative HPLC to give 15 undescribed and 5 known lanostane-type C_{31} triterpenoid derivatives (Fig. 1).

Compound **1** was isolated as a white powder. Its molecular formula $C_{37}H_{58}O_9$ was determined from the ^{13}C NMR (Table 1) and HRESIMS [*m/z* 669.3964 ([M+Na]$^+$)] data. The IR spectrum indicated the presence of hydroxy (3441 cm^{-1}) and carbonyl (1732, 1705 cm^{-1}) groups in **1**. Eight methyls, three oxygenated methines, a 1,1-di-substituted double bond, a tetra-substituted double bond, and three carbonyls were recognized from the NMR spectroscopic data of **1** (Table 1), together with ten aliphatic methylenes, four aliphatic methines, and five quaternary carbons. A 3-hydroxy-3-methylglutaryloxy moiety [$HOOCCH_2C(CH_3)(OH)CH_2C(O)O$-, moiety A, Fig. 2A] was established from the following HMBC correlations: H-2′/C-1′, C-3′, C-6′; H-4′/C-3′, C-5′, C-6′; and H-6′/C-2′, C-3′, C-4' (Fig. 2A). The remaining NMR spectroscopic data were similar to those of 3α-acetylpolyporenic acid A (**10**, Fig. 1), an acetylated lanostane-type C_{31} tetracyclic triterpenoid acid (Wangun et al., 2004). Comparison of the NMR spectroscopic data of **1** (Table 1) with those of **10** suggested that another oxygenated methine rather than a methylene group presented in **1** (Fig. 1). The HMBC correlations of H-6/C-5, C-7, C-8, C-10 and H-12/C-9, C-11, C-13, C-14, C-18 suggested that two hydroxy groups were located at C-6 and C-12, respectively (Fig. 2A). The moiety A was located at C-3 according to the key HMBC correlation of H-3/C-1′ (Fig. 2A). Comparing the coupling constants of H-3 and H-2 with those from the similar structures (Jiang et al., 1977; Kamo et al., 2003; Lai et al., 2016; Li et al., 1993; Tohtahon et al., 2017; Wang et al., 2003), the double doublets of H-3 (δ_H 4.45, 1 H, dd, $J = 11.6, 4.3$ Hz) in **1** suggested a β-orientation of moiety A, which was confirmed by the NOESY correlation of H-3 and the α-orientated H-5 in all lanostane-type triterpenoids (Jiang et al., 1977; Kamo et al., 2003; Lai et al., 2016; Li et al., 1993; Tohtahon et al., 2017). The relative configurations of **1**

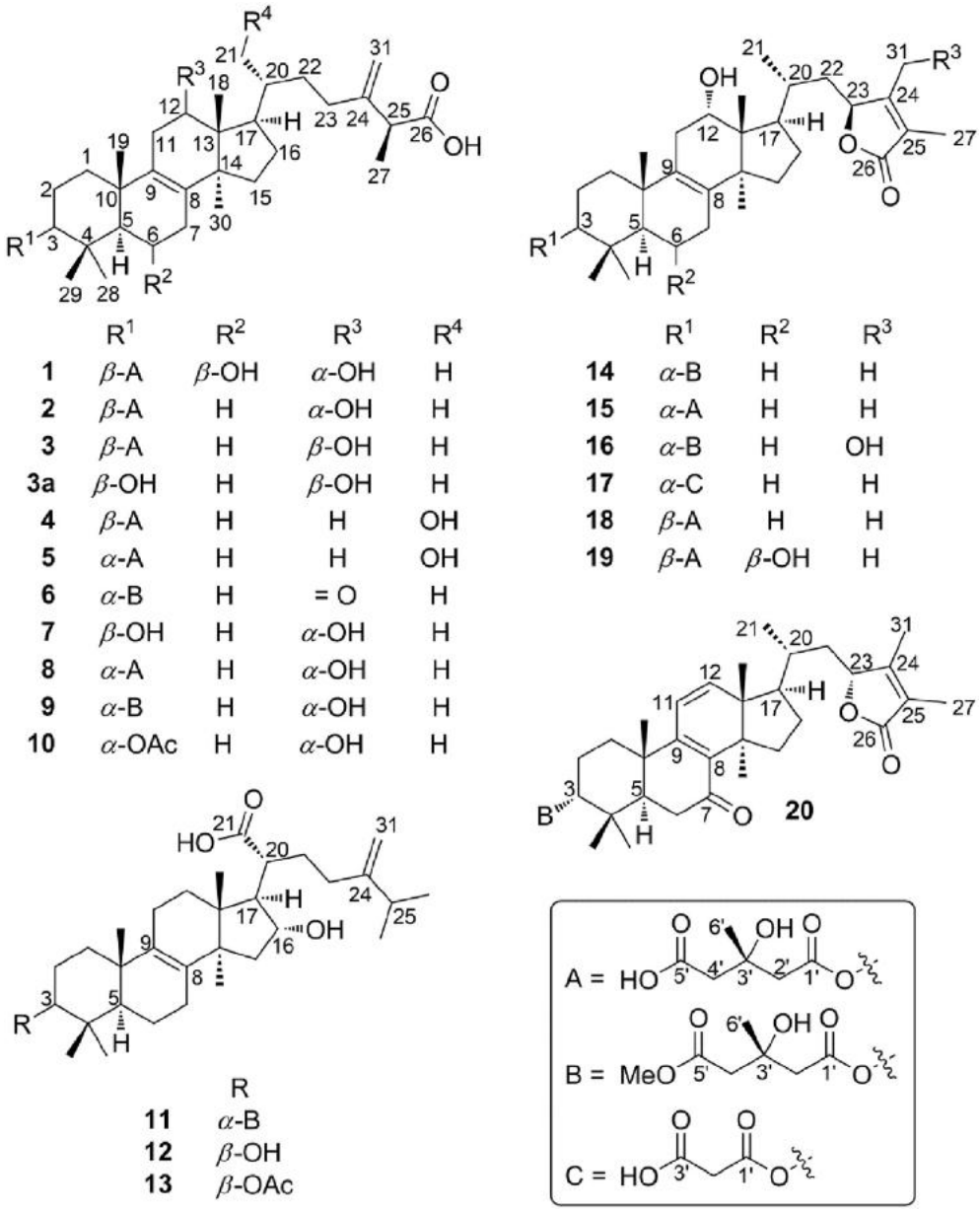

Fig. 1. Structures of the lanostane-type C_{31} triterpenoid derivatives from *Fomitopsis palustris*.

Table 1
NMR Spectroscopic Data for Compounds **1**, **11**, **14**, and **20**.

Position	**1**[a]		**11**[b]		**14**[c]		**20**[d]	
	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type
1	1.74, m	38.72, CH_2	1.88, m	31.84, CH_2	1.56,m	31.89, CH_2	1.74, m	31.82, CH_2
	1.30, m		1.48, m				1.50, m	
2	1.76, m	25.36, CH_2	1.87, m	24.11, CH_2	1.97, m	24.19, CH_2	2.00, m	23.62, CH_2
			1.64, m		1.69, m		1.91, m	
3	4.45, dd (11.6, 4.3)	82.54, CH	4.69, brs	79.60, CH	4.69, brs	79.74, CH	4.96, brs	77.53, CH
4		39.56, C		37.65, C		37.76, C		37.18, C
5	1.09, m	54.15, CH	1.55, m	46,67, CH	1.57, m	46.85, CH	2.36, d (3.8)	46.17, CH
6	4.48, d (5.5)	65.68, CH	1.61, m	18.96, CH_2	1.68, m	19.10, CH_2	2.55, m	36.81, CH_2
			1.51, m		1.58, m			
7	2.43, m	38.33, CH_2	2.02, m	27.10, CH_2	2.08, m	27.11, CH_2		199.18,C
	2.10, m							
8		133.31, C		135.65, C		136.03, C		136.56, C
9		135.53, C		135.61, C		134.47, C		158.57, C
10		37.67, C		37.91, C		37.89, C		38.83, C
11	2.69, m	34.17, CH_2	2.08, m	21.44, CH_2	2.66, m	34.36, CH_2	6.18, d (9.8)	122.68, CH
	2.09, m				2.10, m			
12	4.01, d (7.9)	73.76, CH	1.83, m	30.12, CH	4.00, d (8.0)	73.50, CH	6.62, d (9.8)	147.52, CH
			1.53, m					
13		50.47, C		46.82, C		50.68, C		48.45, C
14		50.68, C		49.25, C		50.75, C		49.33, C
15	1.76, m	33.41, CH_2	2.16, m	43.68, CH_2	1.72, m	33.23, CH_2	1.88, m	31.24, CH_2
	1.18, m		1.33, d (13.3)		1.19, m		1.29, m	
16	2.06, m	28.96, CH_2	4.12, t (7.1)	77.58, CH	2.10, m	29.07, CH_2	2.37, m	30.15, CH_2
							2.15, m	
17	2.70, m	43.83, CH	2.19, m	57.28, CH	2.36, d (8.6)	44.13, CH	2.36, dd (13.7,3.9)	44.15, CH
18	0.71, s	17.10, CH_3	0.82, s	17.85, CH_3	0.70, s	16.87, CH_3	0.94, s	14.97, CH_3
19	1.379, s	20.45, CH_3	0.96, s	19.42, CH_3	1.05, s	19.29, CH_3	1.17, s	20.33, CH_3
20	1.44, m	37.55, CH	2.47, t (10.7)	49.06, CH	1.71, m	36.66, CH	1.71, m	36.40, CH
21	1.05, d (6.5)	18.00, CH_3		180.57, C	1.19, d (6.6)	19.39, CH_3	1.18, d (6.5)	20.09, CH_3
22	1.65, m	35.56, CH_2	2.12, m	31.89, CH_2	2.03, m	40.23, CH_2	1.86, m	39.55, CH_2
	1.28, m		1.88, m		1.29, m			
23	2.24, m	32.87, CH_2	2.12, m	33.50, CH_2	4.90, d (8.2)	85.21, CH	4.80, d (9.2)	83.40, CH
	2.04, m		2.03, m					
24		150.52, C		156.67, C		163.47, C		160.91, C
25	3.13, q (7.0)	46.55, CH	2.22, m	34.88, CH		123.47, C		123.29, C
26		178.40, C	1.00, d (5.8)	22.39, CH_3		177.03, C		174.73, C
27	1.26, d (7.0)	16.85, CH_3	0.98, d (5.7)	22.27, CH_3	1.80, s	8.33, CH_3	1.79, m	8.86, CH_3
28	0.99, s	28.54, CH_3	0.88, s	28.39, CH_3	0.90, s	28.37, CH_3	0.92, s	28.39, CH_3
29	1.29,s	18.78, CH_3	0.90, s	22.28, CH_3	0.98, s	22.27, CH_3	0.92, s	22.30, CH_3
30	1.10, s	25.05, CH_3	1.15, s	25.60, CH_3	1.15, s	25.26, CH_3	1.11, s	25.28, CH_3
31	4.93, s	111.13, CH_2	4.77, s	107.38, CH_2	2.04, s	12.36, CH_3	1.85, s	12.29, CH_3
	4.90, s		4.75, s					
1′		172.63, C		172.20, C		172.32, C		171.44, C
2′	2.74, d (14.5)	46.60, CH_2	2.78, d (14.7)	46.57, CH_2	2.75, d (14.7)	46.42, CH_2	3.09, d (14.5)	46.49, CH_2
	2.67, d (14.5)		2.72, d (14.7)		2.69, d (14.7)		3.01, d (14.5)	
3′		70.72, C		70.68, C		70.77, C		70.21, C
4′	2.66, s	46.02, CH_2	2.74, s	46.03, CH_2	2.72, s	45.90, CH_2	3.05, d (14.7)	46.74, CH_2
							2.96, d (14.7)	
5′		175.02, C		172.94, C		173.04, C		172.24, C
6′	1.381, s	27.73, CH_3	1.41, s	28.20, CH_3	1.39, s	28.03, CH_3	1.69, s	28.86, CH_3
5′-OMe			3.64, s	51.96, CH_3	3.68, s	52.01, CH_3	3.63, s	51.65, CH_3

[a] Recorded at 600 MHz for ^1H and 100 MHz for ^{13}C in Methanol-d_4.
[b] Recorded at 400 MHz for ^1H and 150 MHz for ^{13}C in Methanol-d_4: Pyridine-d_5 (19:1).
[c] Recorded at400 MHz for ^1H and 100 MHz for ^{13}C in Methanol-d_4.
[d] Recorded at 600 MHz for ^1H and 150 MHz for ^{13}C in Pyridine-d_5. Chemical shifts and coupling constants (in parentheses) are given in ppm and Hz, respectively. s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet.

were deduced from the NOESY correlations (Fig. 2B) and confirmed by the X-ray crystallographic analysis (Fig. 2C). The absolute configuration of C-3′ was determined to be *S* on the basis of the absolute configurations of the well-known lanostane-type triterpenoids (Hill and Connolly, 2017; Nes, 2011), which is in excellent agreement with the mevalonate biosynthetic origin of moiety A (Pérez-Gil and Rodríguez-Concepción, 2013; Schwab and Wüst, 2015). Thus compound **1** was determined as (3*S*,6*S*,12*S*,3′*S*)-6,12-dihydroxy-3-(3′-hydroxy-3′-methylglutaryloxy)-24-methyl lanosta-8,24(31)-dien-26-oic acid and was named palustrisoic acid A (Fig. 1).

Compound **2** (palustrisoic acid B) was obtained as a white powder. The molecular formula of **2** was determined to be $C_{37}H_{58}O_8$ from the ^{13}C NMR (Table 2) and HRESIMS [*m/z* 653.4038 ($[M+Na]^+$)] data, indicating that **2** has a 16 amu less than that of **1**. Comparison of the NMR spectroscopic data of **2** (Table 2) with those of **1** (Table 1) suggested the presence of moiety A. The NMR spectroscopic data of **2** were similar to those of **1** and (25*S*,3′*S*)-(+)-12α-hydroxy-3α- [3′-hydroxy-3′-methylglutaryloxy]-24-methyllanosta-8,24 (31)-dien-26-oic acid (**8**) (Kamo et al., 2003). The planar structure of **2** was established from 2D NMR experiments (Fig. S1A in Supplementary Data). The coupling constants of H-3 and H-2 (Table 2) and the NOESY correlation of H-3 and the α-orientated H-5 in **2** (Fig. S1B in Supplementary Data) suggested a β-

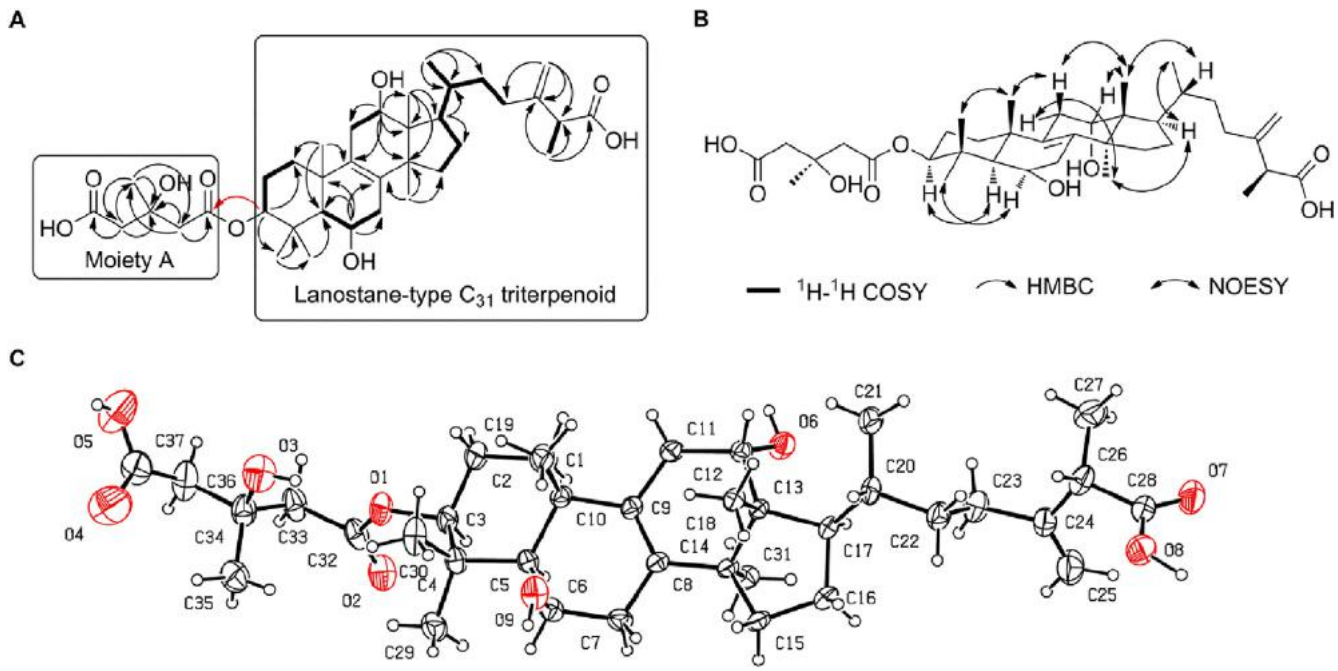

Fig. 2. ^{1}H-^{1}H COSY and Key HMBC correlations (**A**), selected NOESY correlations (**B**), and ORTEP drawing (**C**) of compound **1**.

orientation of moiety A. The relative configuration of the lanostane-type C_{31} triterpenoid moiety of **2** was established by the NOESY correlations (Fig. S1B in Supplementary Data). The absolute configuration of moiety A was determined by comparison of the NMR spectroscopic data of **2** with those of **1** and in view of their identical biosynthetic origin. Thus, compound **2** is the C-3 epimer of **8** (Fig. 1).

Compound **3**, palustrisoic acid C, was shown to have the molecular formula $C_{37}H_{58}O_8$ by the ^{13}C NMR (Table 2) and HRESIMS [*m/z* 653.4043 ([M+Na]$^+$)] data. The NMR spectroscopic data of **3** were similar to those of **2** (Table 2). The difference is that the ^{1}H NMR signal for H-12 of **2** is at δ_H 4.01 (d, $J = 7.8$ Hz) whereas H-12 in **3** is at δ_H 4.06 (t, $J = 7.8$ Hz), indicating different relative configuration of C-12. The key NOEY correlations of H-12/H-30/H-17 confirmed the β-orientation of 12-OH in compound **3** (Fig. S1B in Supplementary Data). Thus, compound **3** was determined as the C-12 epimer of **2** (Fig. 1).

The molecular formula $C_{37}H_{58}O_8$ was established for compound **4**, palustrisoic acid D, by the ^{13}C NMR (Table 2) and HRESIMS [*m/z* 653.4007 ([M+Na]$^+$)] data. The NMR spectroscopic data of **4** are similar to those of **1**–**3** (Tables 1 and 2). The ^{1}H-^{1}H COSY correlations of H-21/H-20/H-22/H-23 and H-20/H-17/H-16/H-15 and the HMBC correlations of H-21/C-17, C-20, C-22 showed the presence of a hydroxymethyl group in **4** rather than an oxygenated methine group in **3** (Fig. S1A in Supplementary Data). The planar structure and relative configurations of **4** were established from 2D NMR experiments (Fig. S1 in Supplementary Data). The absolute configurations of **4** were established by comparison of its NMR spectroscopic data with those of **1**–**3** and in view of their identical biosynthetic origin.

Compound **5**, palustrisoic acid E, has the molecular formula of $C_{37}H_{58}O_8$ by the ^{13}C NMR (Table 2) and HRESIMS [*m/z* 653.4024 ([M+Na]$^+$)] data. The NMR spectroscopic data of **5** were similar to those of **4** (Table 2). The ^{1}H NMR signal for H-3 in **5** is at δ_H 4.70 (brs) rather than that for H-3 (δ_H 4.51, dd, $J = 8.4$, 6.0 Hz) in **4** (Table 2). Thus **5** is the C-3 epimer of **4** (Fig. 1), which was supported by the NMR spectroscopic data (Fig. S1 in Supplementary Data and Table 2).

Compound **6**, palustrisoic acid F, was obtained as a white powder. Its molecular formula $C_{38}H_{58}O_8$ was established by the ^{13}C NMR (Table 2) and HRESIMS [*m/z* 665.4042 ([M+Na]$^+$)] data. The NMR spectroscopic data of **6** (Table 2) were similar to those of (25S)-(+)-12α-hydroxy-3α-[(3*S*)-3-hydroxy-4-methoxycarbonyl-3-methylbutyryloxy]-24-methyllanosta-8,24 (31)-dien-26-oic acid (**9**) (Kamo et al., 2003). The difference between compounds **6** and **9** was that an oxygenated methine group of **9** was replaced by a carbonyl group to form **6** (Fig. 1). The key HMBC correlations of H-18/C-12, C-13, C-14, C-17 suggested the presence of a C-12 keto group (Fig. S1A in Supplementary Data). The methoxy group was located at C-5′ to give moiety B (3-hydroxy-4-methoxycarbonyl-3-methylbutyryloxy), which was confirmed by the key HMBC correlations (Fig. S1A in Supplementary Data). The broad singlet of H-3 suggested an α-orientation of moiety B (Li et al., 2016; Rösecke and König, 2000). The absolute configurations of **6** were established by using the identical methods described for **2**–**5**.

Compound **7** was obtained as a white powder. Its molecular formula $C_{31}H_{50}O_4$ was established by the ^{13}C NMR (Table 3) and HRESIMS [*m/z* 509.3614 ([M+Na]$^+$)] data. The NMR spectroscopic data of **7** (Table 3) were similar to those of **2** without moiety A (Table 2). The planar structure and relative configurations of **7** were established from NMR spectroscopic data (Fig. S1 in Supplementary Data). The exact structure search using SciFinder (https://scifinder.cas.org) suggested that fomefficinic acid G (Wu et al., 2009) has the identical structure and relative configurations to those of **7**. However, their corresponding NMR spectroscopic data were not identical (Table 3), which suggested that the relative configurations of C-3 and C-12 in **7** and fomefficinic acid G were different. The NMR spectroscopic data of both **7** and fomefficinic acid G were different with those of polyporenic acid A, the known 3α,12α-dihydroxy-24-methyllanosta-8,24 (31)-dien-26-oic acid, whose structure was confirmed by X-ray crystallography of its methyl ester (King et al., 1984). The coupling constants of H-3 and H-2 of both **7** and fomefficinic acid G (Table 3) suggested a β-orientation of 3-OH, which are different from the known 3α-OH lanostane-type C_{31} triterpenoid acid (Hill and Connolly, 2017; Li et al., 1993; Kamo et al., 2003; King et al., 1984; Lai et al., 2016; Nes, 2011; Tohtahon

Table 2
NMR Spectroscopic Data for Compounds **2**–**6**.

Position	**2**[a]		**3**[a]		**4**[a]		**5**[b]		**6**[a]	
	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type
1	1.67, m	35.61, CH_2	1.81, m	36.61, CH_2	1.69, m	36.50, CH_2	1.66, m	31.82, CH_2	1.49, m	31.89, CH_2
	1.32, m		1.31, m		1.29, m		1.26, m		1.46, m	
2	1.72, m	25.12, CH_2	1.72, m	25.15, CH_2	1.70, m	25.17, CH_2	1.95, m	24.21, CH_2	2.18, m	26.96, CH_2
							1.65, m		2.10, m	
3	4.52, dd (11.2, 5.8)	82.46, CH	4.51, dd (8.8,4.1)	82.42, CH	4.51, dd (8.4,6.0)	82.55, CH	4.70, brs	79.90, CH	4.72, brs	79.55, CH
4		38.79, C		38.81, C		38.83, C		37.76, C		37.74, C
5	1.20, m	52.03, CH	1.14, m	51.98, CH	1.18, m	51.99, CH	1.54, m	46,78, CH	1.59, m	46,74, CH
6	1.77, m	19.23, CH_2	1.73, m	19.21, CH_2	1.74, m	19.25, CH_2	1.66, m	19.13, CH_2	1.72, m	18.87, CH_2
	1.61, m		1.58, m		1.59, m		1.55, m		1.58, m	
7	2.12, m	27.45, CH_2	2.09, m	27.20, CH_2	2.09, m	27.49, CH_2	2.09, m	27.14, CH_2	1.98, m	24.02, CH_2
									1.68, m	
8		136.39, C		135.55, C		135.69, C		135.54, C		136.54, C
9		134.01, C		136.82, C		135.83, C		136.02, C		136.45, C
10		37.92, C		37.95, C		38.13, C		38.05, C		37.92, C
11	2.64, m	34.37, CH_2	2.52, m	34.67, CH_2	2.09, m	22.10, CH_2	2.08, m	22.02, CH_2	3.02,brd	40.87, CH_2
	2.12, m		1.82, m						2.66, m	
12	4.01 d (7.8)	73.59, CH	4.06, t (7.8)	73.72, CH	1.86, m	31.84, CH_2	1.85, m	31.68, CH_2		215.07, C
					1.67, m		1.67, m			
13		50.68, C		50.23, C		45.55, C		45.55, C		60.77, C
14		50.57, C		53.21, C		51.05, C		51.10, C		55.73, C
15	1.73, m	33.25, CH_2	1.75, m	32.36, CH_2	1.67, m	31.67, CH_2	1.66, m	32.05, CH_2	1.93, m	31.50, CH_2
	1.20, m		1.21, m		1.25, m		1.25, m		1.38, m	
16	2.05, m	28.95, CH_2	2.03, m	25.79, CH_2	1.98, m	28.48, CH_2	1.98, m	28.46, CH_2	2.07, m	28.75, CH_2
	1.36, m		1.34, m		1.53, m		1.53, m			
17	2.25, q (9.7)	43.78, CH	1.91, m	51.98, CH	1.89, m	45.55, CH	1.89, m	45.68, CH	2.18, m	43.65, CH
18	0.67, s	17.02, CH_3	0.74, s	10.82, CH_3	0.76, s	16.67, CH_3	0.79, s	16.65, CH_3	1.13, s	13.13, CH_3
19	1.05, s	19.49, CH_3	1.07, s	19.64, CH_3	1.05, s	19.67, CH_3	1.07, s	19.48, CH_3	1.17, s	19.39, CH_3
20	1.45, m	37.59, CH	1.75, m	35.14, CH	1.53, m	44.02, CH	1.53, m	44.03, CH	1.33, m	37.68, CH
21	1.06, d (6.9)	17.99, CH_3	1.04, d (7.0)	21.98, CH_3	3.73, dd (11.0,2.9)	62.74, CH_2	3.75, dd (11.0,2.8)	62.80, CH_2	0.90,d (7.5)	19.64, CH_3
					3.52, dd (11.0,5.2)		3.55, dd (11.0,5.0)			
22	1.83, m	36.29, CH_2	1.84, m	34.88, CH_2	1.67, m	28.95, CH_2	1.67, m	28.99, CH_2	1.67, m	35.36, CH_2
	1.33, m		1.20, m		1.53, m		1.53, m		1.28, m	
23	2.25, m	32.88, CH_2	2.20, m	34.13, CH_2	2.18, m	32.40, CH_2	2.20, m	32.44, CH_2	2.23, m	33.02, CH_2
	2.07, m		2.12, m		2.10, m		2.09, m		2.04, m	
24		150.56, C		150.76, C		150.59, C		150.70, C		150.60, C
25	3.15, q (7.0)	46.57, CH	3.13, q (7.0)	46.68, CH	3.15, q (7.0)	46.60, CH	3.16, q (7.0)	46.83, CH	3.12, q (7.0)	46.93, CH
26		178.36, C		178.45, C		178.44, C		178.82, C		178.99, C
27	1.28, d (7.0)	16.87, CH_3	1.26, d (7.0)	16.86, CH_3	1.26, d (7.1)	16.91, CH_3	1.28, d (7.0)	16.93, CH_3	1.26, d (7.0)	16.92, CH_3
28	0.94, s	28.54, CH_3	0.91, s	28.54, C H_3	0.91, s	28.55, CH_3	0.92, s	28.40, CH_3	0.94, s	28.32, CH_3
29	0.95, s	17.15, CH_3	0.93, s	17.12, C H_3	0.93, s	17.13, CH_3	0.98, s	22.29, CH_3	1.01, s	22.17, CH_3
30	1.13, s	25.31, CH_3	0.94, s	24.59, CH_3	0.94, s	24.83, CH_3	0.98, s	24.84, CH_3	0.83, s	24.46, CH_3
31	4.94, s	111.12, CH_2	4.92, s	111.06, CH_2	4.95, s	111.19, CH_2	4.95, s	111.10, CH_2	4.95, s	111.03, CH_2
	4.91, s		4.88, s		4.94, s		4.91, s		4.91, s	
1′		172.59, C		172.62, C		172.65, C		172.51, C		172.24, C
2′	2.75, d (14.8)	46.58, CH_2	2.73, d (14.3)	46.63, CH_2	2.75, d (14.6)	46.60, CH_2	2.71, s	46.67, CH_2	2.72, s	46.44, CH_2
	2.67, d (14.8)		2.65, d (14.3)		2.67, d (14.6)					
3′		70.71, C		70.73, C		70.72, C		70.84, C		70.76, C
4′	2.67,s	46.02, CH_2	2.64, s	46.12, CH_2	2.68, s	46.01, CH_2	2.65, m	46.36, CH_2	2.71, s	45.93, CH_2
									2.69, m	
5′		175.02, C		175.29, C		174.92, C		176.20, C		173.02, C
6′	1.40, s	27.73, CH_3	1.37, s	27.74, CH_3	1.38, s	27.74, CH_3	1.38, s	27.88, CH_3	1.38, s	28.03, CH_3
5′-OMe									3.67, s	52.00, CH_3

[a] Recorded at 400 MHz for ^{1}H and 100 MHz for ^{13}C in Methanol-d_4.
[b] Recorded at 400 MHz for ^{1}H and 150 MHz for ^{13}C in Methanol-d_4. Chemical shifts and coupling constants (in parentheses) are given in ppm and Hz, respectively.

et al., 2017). The NOESY correlation of H-3 and H-5 in **7** confirmed this conclusion (Fig. S1B in Supplementary Data). The relative configuration of C-12 in fomefficinic acid G was established from the NOESY correlation of H-12/H-19 and the coupling constants of H-12 and H-11 (Sakurai et al., 1986, 1990; Wu et al., 2009). Wu et al. calculated the coupling constants of H-12 and H-11 on the basis of the Karplus equitation and the torsion angle of C-9-C-11-C-12-C-13 (- 14.4°) of beesioside II, the known 9,19-cyclolanstanol glycoside, whose structure was confirmed by X-ray crystallography of the diacetate of the aglycone of beesioside II (Sakurai et al., 1986, 1990; Wu et al., 2009). The authors suggested that if 12-OH is α-orientated, the coupling constants of H-12 and H-11 should be 7.7 and 4.4 Hz. If 12-OH is β-orientated, the coupling constants of H-12 and H-11 should be 7.7 and 0.4 Hz (Wu et al., 2009). The ^{1}H NMR signal of H-12 was appeared as triplet-like (7.5 Hz), which suggested the α-orientation of 12-OH (Wu et al., 2009). However, the torsion angle of C-9-C-11-C-12-C-13 is - 4.8° in 12α-OH lanostane-type C_{31} triterpenoid based on the X-ray crystallographic structure of **1**. The NOESY correlation of H-12/H-18/H-20 rather than H-12/H-19 was observed in **7** and the ^{1}H NMR signal of H-12 is resonated in doublet (Fig. S1B in Supplementary Data and Table 3), which is in agreement with those of the 12α-OH lanostane-type C_{31} triterpenoids. We concluded that **7** is 3β,12α-dihydroxy- and fomefficinic acid G should be 3β,12β-dihydroxy-substituted 24-methyllanosta-8,24 (31)-dien-26-oic acid. To support this structure revision of fomefficinic aid G, compounds **2** and **3** that have the identical lanostane-type C_{31} triterpenoid moiety to **7** and fomefficinic acid G, respectively, were subjected to alkaline hydrolysis. The NMR

Table 3
NMR Spectroscopic Data for Compounds **7**, **3a**, and Fomefficinic acid.

Position	**7**[a]		**3a**[a]		Fomefficinic acid[b]	
	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type
1	1.90, m	35.49, CH_2	1.71, m	36.51, CH_2	1.72, m	36.3, CH_2
	1.52, m		1.23, m		1.26, m	
2	1.89, m	29.08, CH_2	1.78, m	31.99, CH_2	1.80, m	31.7, CH_2
			1.21, m		1.22, m	
3	3.48, dd (11.0, 5.1)	78.44, CH	3.46, dd (9.7, 6.3)	78.37, CH	3.44, dd (7, 9)	78.0, CH
4		39.89, C		39.74, C		39.5, C
5	1.27, m	51.43, CH	1.17, m	51.22, CH	1.45, m	51.0, CH
6	1.81, m	19.15, CH_2	1.16, m	18.99, CH_2	1.77, m	18.7, CH_2
	1.62, m		1.07, m		1.08, m	
7	2.16, m	27.25, CH_2	1.86, m	28.89, CH_2	1.86, m	28.7, CH_2
			1.23, m		1.22, m	
8		135.45, C		136.94, C		136.7, C
9		134.25, C		134.34, C		134.1, C
10		37.28, C		37.50, C		37.3, C
11	2.78, dd (18.8, 7.6)	34.97, CH_2	2.77, m	35.27, CH_2	2.76, m	35.1, CH_2
	2.45, dt (18.7, 3.3)		2.21, m		2.21, m	
12	4.28, d (7.6)	72.49, CH	4.42, t (7.7)	72.30, CH	4.41, t (7.5)	72.0, CH
13		50.43, C		52.59, C		52.3, C
14		50.37, C		49.95, C		49.7, C
15	1.78, m	32.91, CH_2	1.96, m	34.62, CH_2	1.96, m	34.4, CH_2
	1.27, m		1.64, m		1.64, m	
16	2.16, m	28.70, CH_2	2.06, m	25.90, CH_2	1.46, m	25.7, CH_2
	1.45, m				2.06, m	
17	2.70, q (9.7)	43.55, CH	2.10, m	51.66, CH	2.08, m	51.4, CH
18	0.79, s	17.15, CH_3	1.04, s	11.09, CH_3	1.04, s	10.9, CH_3
19	1.10, s	19.66, CH_3	1.02, s	24.56, CH_3	1.02, s	24.3, CH_3
20	1.57, m	37.66, CH	2.09, m	26.84, CH	2.10, m	26.6, CH
21	1.32, d (6.4)	18.28, CH_3	1.35, d (5.8)	22.12, CH_3	1.34, d (6.5)	21.9, CH_3
22	1.76, m	36.35, CH_2	2.10, m	34.76, CH_2	2.17, m	34.5, CH_2
	1.31, m		1.45, m		1.47, m	
23	2.60, m	33.04, CH_2	2.57, m	33.82, CH_2	2.58, m	33.6, CH_2
	2.35, m		2.37, m		2.36, m	
24		150.57, C		151.02, C		150.8, C
25	3.51, q (7.1)	46.96, CH	3.52, q (7.2)	46.89, CH	3.55, q (7, 14)	46.5, CH
26		177.22, C		177.67, C		176.9, C
27	1.55, d (7.0)	17.45, CH_3	1.52, d (7.1)	17.33, CH_3	1.52, d (7)	17.1, CH_3
28	1.28, s	28.99, CH_3	1.07, s	16.60, CH_3	1.07, s	16.4, CH_3
29	1.12, s	16.72, CH_3	1.24, s	28.89, CH_3	1.24, s	28.6, CH_3
30	1.50, s	25.61, CH_3	1.08, s	19.68, CH_3	1.08, s	19.5, CH_3
31	5.25, s	110.65, CH_2	5.24, s	110.60, CH_2	5.24, s	110.4, CH_2
	5.14, s		5.13, s		5.13, s	

[a] Recorded at 600 MHz for ^{1}H and 150 MHz for ^{13}C in Pyridine-d_5.
[b] Recorded at 500 MHz for ^{1}H and 125 MHz for ^{13}C in Pyridine-d_5 and copied from Wu, X.; Yang, J.; Yan, M. *Chem. Pharm. Bull.* **2009**, *57*, 195–197. Chemical shifts and coupling constants (in parentheses) are given in ppm and Hz, respectively.

spectroscopic data, HRESIMS, and optical rotations of the triterpenoid acid from **2** were identical to those of **7**. The aforementioned data of the hydrolysis product **3a** (Table 3) from **3** were identical to those of fomefficinic acid G (Table 3). Thus, for the lanostane-type C_{31} triterpenoids, the doublet of H-12 suggested an α-orientation of 12-OH and the triplet-like of H-12 revealed a β-orientation of 12-OH. Compound **7** was determined as 3β,12α-dihydroxy-24-methyllanosta-8,24(31)-dien-26-oic acid and was named palustrisoic acid G (Fig. 1).

Compound **11** was obtained as a white powder. Its molecular formula $C_{38}H_{60}O_8$ was determined from the ^{13}C NMR (Table 1) and HRESIMS [*m/z* 667.4209 ([M+Na]$^+$)] data. The IR spectrum indicated the presence of hydroxy (3432 cm^{-1}) and carbonyl (1733, 1715 cm $^{-1}$) groups. The NMR spectroscopic data of **11** (Table 1) showed the presence of moiety B (Fig. 3A). The remaining NMR spectroscopic data of **11** (Table 1) were similar to those of polyporenic acid B (**12**) (Shibata et al., 1958) and pachymic acid (**13**) (Keller et al., 1996; Shibata et al., 1958; Shingu et al., 1992). The moiety B was located at C-3 in view of the key HMBC correlations of H-3/C-1′ (Fig. 3A). The broad singlet of H-3 suggested an α-orientation of moiety B (Li et al., 2016; Rösecke and König, 2000). The NOESY correlations of H-16/H-18/H-20 indicated the α-orientation of 16-OH and 20-COOH (Fig. 3B). The relative configurations of **11** established from the NOESY correlations (Fig. 3B) were confirmed by the X-ray crystallographic analysis (Fig. 3C). The absolute configuration of C-3′ was determined to be *S* by using the methods described for **1**. Thus, compound **11** was determined to be (3R,16R,3′*S*)-16-hydroxy-3-[(3*S*)-3-hydroxy-4-methoxycarbonyl-3-methyl-butaryloxy]-24-methyllanosta-8,24 (31)-dien-21-oic acid and has been named palustrisoic acid H (Fig. 1).

The molecular formula $C_{38}H_{58}O_8$ for compound **14** was established from the ^{13}C NMR (Table 1) and HRESIMS [*m/z* 665.4038 ([M+Na]$^+$)] data. The IR spectrum indicated the presence of hydroxy (3451 cm^{-1}) and carbonyl (1735, 1690 cm $^{-1}$) groups in **14**. The NMR spectroscopic data of **14** (Table 1) showed the presence of moiety B (Fig. 4A). An α,β-unsaturated γ-lactone was established from the key HMBC correlations of H-23/C-26; H-27/C-24, C-25, C-26; and H-31/C-23, C-24, C-25 (Fig. 4A). The remaining NMR spectroscopic data of **14** (Table 1) were similar to those of compounds **2** and **3** (Table 2), indicating the presence of similar lanostane-type triterpenoid framework. The moiety B was located at C-3 on the basis of the key HMBC correlation of H-3/C-1′ (Fig. 4A). A hydroxy group was located at C-12 by the HMBC correlations of H-12/C-9, C-11, C-13, C-18 and H-18/C-12, C-13, C-14, C-17 (Fig. 4A).

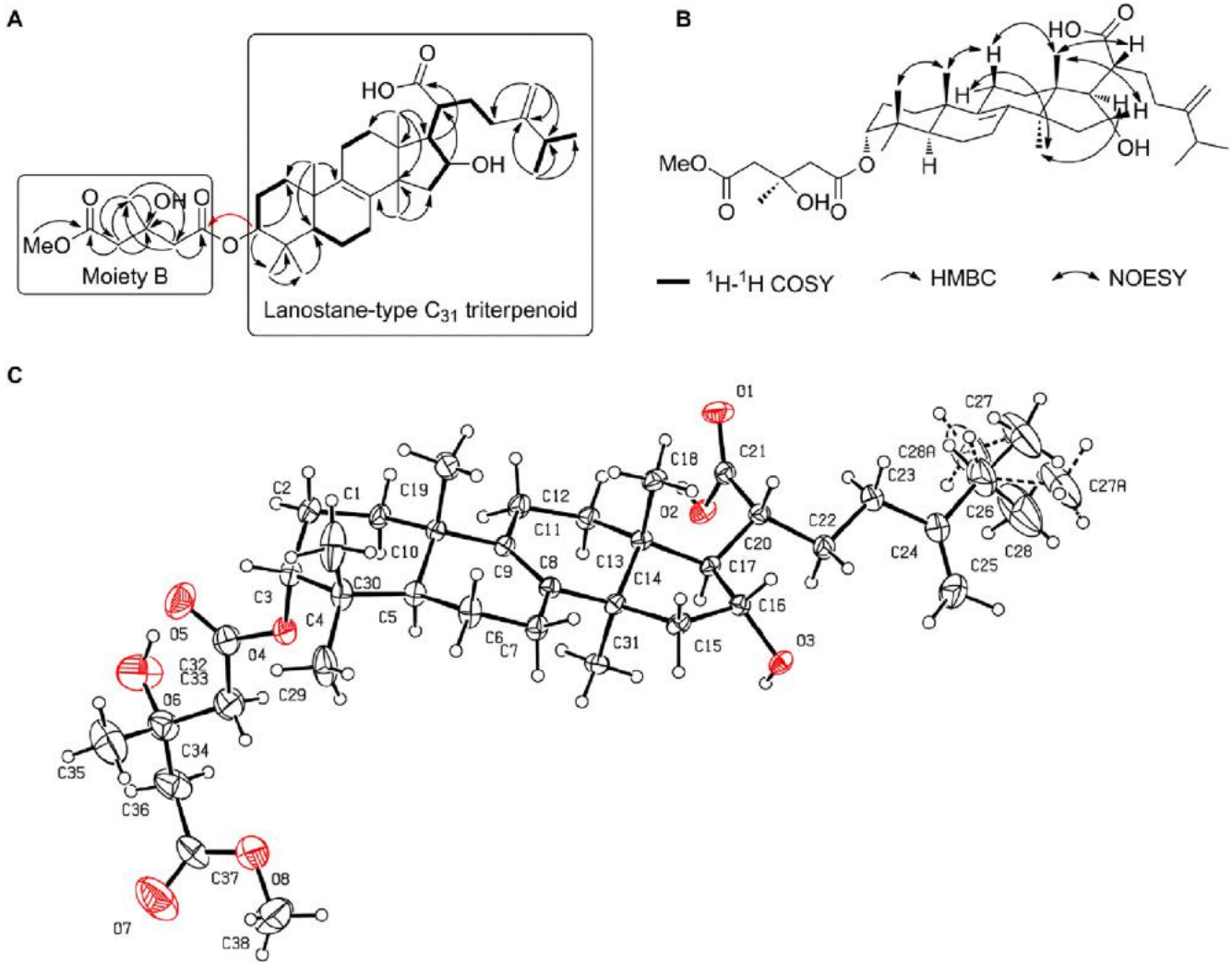

Fig. 3. ^{1}H-^{1}H COSY and key HMBC correlations (**A**), selected NOESY correlations (**B**), and ORTEP drawing (**C**) of compound **11**.

The broad singlet of H-3 at δ_H 4.69 suggested an α-orientation of moiety B (Li et al., 2016; Rösecke and König, 2000). The relative configurations of C-10, C-13, C-14, C-17, and C-21 were established from the NOESY correlations (Fig. 4B). The NMR spectroscopic data of the α,β-unsaturated γ-lactone moiety of **14** were similar to those of the known compound whose structure was confirmed by X-ray crystallography (He et al., 2006), indicating the assignment of 23*S* configuration. According to Klyne's lactone sector rule (Jennings et al., 1965), the absolute configuration of C-23 was also assigned to be *S* in view of the positive Cotton effect at 223 nm (Fig. 4C) (Wang et al., 2003), which is in agreement with those reported (Allen et al., 1971). Thus, **14** was identified as (3*R*,12*S*,23*S*,3′*S*)-12-hydroxy-3-(3′-hydroxy-4′-methoxycarbonyl-3′-methylbutyryloxy)-24-methyllanosta-8,24-dien-26,23-olide and has been given trivial name palustrisolide A (Fig. 1).

Compound **15**, palustrisolide B, was obtained as a white powder. Its molecular formula $C_{37}H_{56}O_8$ was determined from the ^{13}C NMR (Table 4) and HRESIMS [*m/z* 651.3883 ([M+Na]$^+$)] data, suggesting that **15** has a 14 amu less than that of **14**. The similarity of the NMR spectroscopic data of **15** (Table 4) and those of **14** (Table 1) suggested that an absence of a methoxy group in **15**. The moiety A was established from the key HMBC correlation (Fig. S2A in Supplementary Data) and it was located at C-3 by the HMBC correlation of H-3/C-1′ (Fig. S2A in Supplementary Data). The absolute configurations of **15** were determined as described for **14** (Fig. 4C).

The molecular formula $C_{38}H_{58}O_9$ of compound **16**, palustrisolide C, was shown to have a 16 amu more than that of **14** by the ^{13}C NMR (Table 4) and HRESIMS [*m/z* 681.3989 ([M+Na]$^+$)] data. Comparison the NMR spectroscopic data of **16** (Table 4) with those of **14** (Table 1) suggested that a methyl group in **14** was replaced by a hydroxymethyl group in **16**, which was confirmed by the HMBC correlations of H-31/C-23, C-24, C-25 (Fig. S2A in Supplementary Data). Thus, compound **16** is a C-31 hydroxylated product of **14** (Fig. 1).

Compound **17** was isolated as a white powder. The molecular formula $C_{34}H_{50}O_7$ was established from the ^{13}C NMR (Table 4) and HRESIMS [*m/z* 593.3439 ([M+Na]$^+$)] data. Comparison of the NMR spectroscopic data of **17** (Table 4) and **14** (Table 1) suggested that they both have the identical lanostane-type C_{31} triterpenoid moiety. The remaining NMR spectroscopic data of **17** (Table 4) were assembled to be a carboxyacetyloxy group (moiety C) by the HMBC correlations of H-2′/C-1′, C-3′ (Fig. S2A in Supplementary Data) and the molecular composition of **17**. The moiety C was located at C-3 by the key HMBC correlations of H-3/C-1′, C-1, C-4, C-5 (Fig. S2A in Supplementary Data). The relative configurations of **17** were determined by using the aforementioned methods. Thus, compound **17**, palustrisolide D, was determined to be (3*R*,12*S*,23*S*)-3-carboxyacetyloxy-12-hydroxy-24-methyllanosta-8,24-dien-26,23-

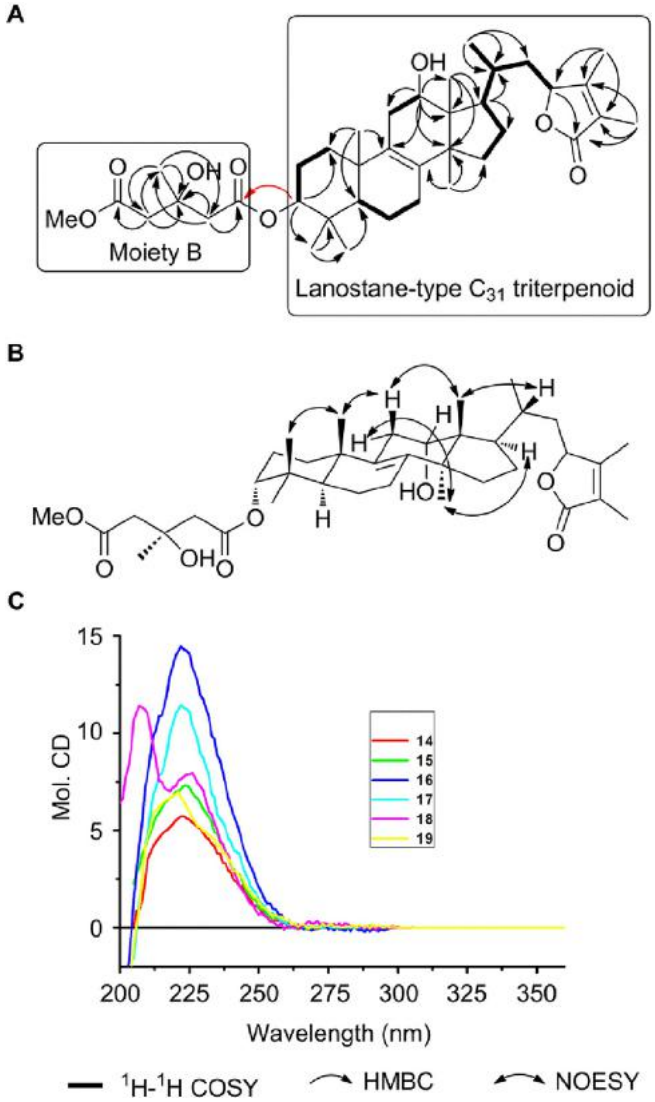

Fig. 4. ^{1}H-^{1}H COSY and key HMBC correlations (**A**) and selected NOESY correlations (**B**) of compound **14**, and ECD spectra of compounds **14**–**19** (**C**).

olide (Fig. 1).

Compound **18**, palustrisolide E, was obtained as a white powder. Its molecular formula $C_{37}H_{56}O_8$ was established by the ^{13}C NMR (Table 4) and HRESIMS [*m/z* 651.3867 ([M+Na]$^+$)] data. The similarity of the NMR spectroscopic data of compounds **18** and **15** (Table 4) suggested that **18** is the C-3 epimer of **15** (Fig. 1). The coupling constants of H-2 and H-3 (11.7 and 4.1 Hz, Table 4) and the NOEY correlations of H-3/H-5 (Fig. S2B in Supplementary Data) revealed that the moiety A is β-orientated.

The molecular formula $C_{37}H_{56}O_9$ of compound **19**, palustrisolide F, was shown to have a 16 amu more than that of **18** by the ^{13}C NMR (Table 4) and HRESIMS [*m/z* 667.3824 ([M+Na]$^+$)] data. Comparison the NMR data of **19** with those of **18** (Table 4) suggested that a methylene group in **18** was replaced by an oxygenated methine in **19**. The hydroxy group was located at C-6 based on the ^{1}H-^{1}H COSY correlation of H-5/H-6/H-7 and the HMBC correlations of H-6/C-5, C-8, C-10 (Fig. S2A in Supplementary Data). The absolute configurations of **19** were determined by using the methods described for **14**.

The molecular formula of **20** was established to be $C_{38}H_{54}O_8$ by the ^{13}C NMR (Table 1) and HRESIMS [*m/z* 661.3705 ([M+Na]$^+$)] data. The moiety B and the α,β-unsaturated γ-lactone fragment were established from the NMR spectroscopic data (Fig. 5A). An $\alpha,\beta,\gamma,\delta$-unsaturated keto moiety was established from 2D NMR experiments (Fig. 5A). The relative configurations of **20** were determined as described for **14**. The negative Cotton effect at 223 nm in the ECD spectrum of **20** (Fig. 5C) suggested the absolute configuration of C-23 is *R* (Allen et al., 1971; Jennings et al., 1965; Tanaka et al., 2004). Thus, compound **20** (Fig. 1) was determined to be (3R,23R,3′*S*)-3-[(3*S*)-3-hydroxy-4-methoxycarbonyl-3-methylbutyryloxy]-24-methyllanosta-7-oxo-8,10,24-trien-26,23-olide (palustrisolide G).

Based on the spectroscopic and physicochemical data, the known compounds were identified as (25*S*,3′*S*)-(+)-12α-hydroxy-3α-[3′-hydroxy-3′-methylglutaryloxy]-24-methyllanosta-8,24 (31)-dien-26-oic acid (**8**, Fig. 1) (Kamo et al., 2003), (25*S*,3′*S*)-(+)-12α-hydroxy-3α-[3′-hydroxy-4′- methoxycarbonyl-3′-methyl-butyryloxy]-24-methyllanosta-8,24 (31)-dien-26-oic acid (**9**, Fig. 1) (Kamo et al., 2003), 3α-acetylpolyporenic acid (**10**, Fig. 1) (Wangun et al., 2004), polyporenic acid B (**12**, Fig. 1) (Shibata et al., 1958), and pachymic acid (**13**, Fig. 1) (Keller et al., 1996; Shibata et al., 1958; Shingu et al., 1992), respectively.

The isolated lanostane-type C_{31} triterpenoid derivatives were subjected to the cytotoxicity assay against the HCT116, A549, and HepG2 cell lines according to the previously reported MTT methods (Yu et al., 2014). The results showed that palustrisolides A (**14**), C (**16**), and G (**20**) exhibited weak cytotoxicity (Table 5). Polyporenic acid B (**12**) showed strong cytotoxicity with IC_{50} values of 8.4–12.2 μM (Table 5).

3. Experimental

3.1. General experimental procedures

X-ray diffraction data were collected from a Bruker Xcalibur E Quest diffractometer. The other general experimental information, see Luo et al., 2016.

3.2. Fungal material

A fresh fruiting body of *F. palustris* was collected from a tree in Chebaling natural reserve (114°15′21″ E, 24°43′40″ N), Shixing County, Guangdong Province, People's Republic of China. The fruiting body was washed three times by sterile water and surface-sterilized three times by sequential immersion in 75% EtOH. Then the surface-sterilized fruiting body was cut into 3–4 mm fragments. The fragments were put into the slants of potato dextrose agar (PDA) and incubated at 25 °C for 7–10 days. The strain was identified by Dr. Xia Luo. The nascent filamentous was transferred to fresh PDA and incubated at 25 °C to prepare the original seed culture. The original seed culture was inoculated into liquid potato dextrose medium and incubated at 25 °C, 200 rpm to prepare the secondary seed culture. The secondary seed culture was inoculated into the bags (18 × 35 cm) containing the fermentation medium (50% of cotton seed hulls, 20% of wood powder, 20% of corncob powder, 8% of rice bran, 1% of $CaCO_3$, and 1% of $CaSO_4$, pH 6.5–7.0) and incubated in dark at 25 °C. Fifty days later, the bags were opened and incubated at 25 °C under light (~2000 lx). Thirty-five days later, the fruiting bodies were harvested.

3.3. Extraction and isolation

The air-dried and powdered fruiting bodies of *F. palustris* (1.56 kg) were extracted with 8 L of 95% EtOH at 70 °C for 4 h, 3 times. EtOH was removed under reduced pressure to yield a crude residue (391 g). The residue was suspended in 1 L of water and

Table 4
NMR Spectroscopic Data for Compounds **15**–**19**.

Position	**15**[a]		**16**[b]		**17**[b]		**18**[b]		**19**[c]	
	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type	δ_H (*J* in Hz)	δ_C, type
1	1.53, m	31.82, CH_2	1.72, m	31.51, CH_2	1.73, m	31.31, CH_2	1.65, m	35.49, CH_2	1.73, m	38.75, CH_2
			1.50, m		1.49, m		1.20, m		1.30, m	
2	1.94, m	24.17, CH_2	1.86, m	24.02, CH_2	1.81, m	23.83, CH_2	1.77, m	24.85, CH_2	1.79, m	25.40, CH_2
	1.66, m		1.76, m						1.72, m	
3	4.71, brs	79.58, CH	4.95	78.48, CH	4.97, t (2.9)	79.29, CH	4.76, dd (11.7,4.1)	81.04, CH	4.44, dd (11.6,4.1)	82.58, CH
4		37.70, C		37.33, C		37.36, C		38.30, C		39.61, C
5	1.58, m	46.75, CH	1.77, m	46.39, CH	1.79, m	46.1, CH	1.24, m	51.12, CH	1.01, s	54.23, CH
6	1.66, m	19.10, CH_2	1.61, m	18.71, CH_2	1.64, m	18.62, CH_2	1.67, m	18.72, CH_2	4.48, d (4.5)	65.70, CH
	1.57, m		1.54, m				1.50, m			
7	2.07, m	27.07, CH_2	2.08, m	26.69, CH_2	2.07, m	26.54, CH_2	2.08, m	26.90, CH_2	2.43, m	38.38, CH_2
									2.10, m	
8		135.97, C		135.38, C		135.28, C		135.36, C		133.31, C
9		134.40, C		134.13, C		134.06, C		133.75, C		133.69, C
10		37.84, C		37.46, C		37.36, C		37.30, C		37.74, C
11	2.64, m	34.29, CH_2	2.72, m	34.88, CH_2	2.71, m	34.77, CH_2	2.68, dd (18.5,7.3)	34.85, CH_2	2.75, m	34.25, CH_2
	2.10, m		2.45, m		2.43, m		2.35, d (18.6)		2.10, m	
12	4.00, d (7.7)	73.44, CH	4.25, d (7.9)	72.76, CH	4.23, d (7.6)	72.19, CH	4.22, d (7.6)	72.14, CH	4.00, d (7.9)	73.69, CH
13		50.65, C		50.40, C		50.28, C		50.31, C		50.55, C
14		50.70, C		50.44, C		50.30, C		50.28, C		50.86, C
15	1.70, m	33.23, CH_2	1.72, m	32.94, CH_2	1.71, m	32.83, CH_2	1.75, m	32.88, CH_2	1.79, m	33.44, CH_2
	1.18, m		1.23, m		1.19, m		1.28, m		1.21, m	
16	2.08, m	29.06, CH_2	2.16, m	28.81, CH_2	2.10, m	28.67, CH_2	2.16, m	28.71, CH_2	2.11, m	29.09, CH_2
			1.63, m		1.37, m		1.41, m		1.42, m	
17	2.36, q (9.7)	44.08, CH	2.75, dd (9.6)	43.93, CH	2.68, m	43.70, H	2.73, q (9.7)	43.73, CH	2.35, q (9.8)	44.21, CH
18	0.69, s	16.89, CH_3	0.76, s	16.98, CH_3	0.74, s	16.87, CH_3	0.73, s	16.88, CH_3	0.74, s	16.99, CH_3
19	1.04, s	19.33, CH_3	1.02, s	19.29, CH_3	1.01, s	19.17, CH_3	1.01, s	19.43, CH_3	1.38, s	20.43, CH_3
20	1.71, m	36.64, CH	1.93, m	36.63, CH	1.75, m	36.20, CH	1.78, m	36.21, CH	1.72, m	36.60, CH
21	1.19, d (6.5)	19.33, CH_3	1.49, d (6.6)	19.58, CH_3	1.40, d (6.6)	19.38, CH_3	1.41, d (6.5)	19.40, CH_3	1.17, d (6.6)	19.33, CH_3
22	2.01, m	40.21, CH_2	2.40, m	40.65, CH_2	1.97, m	39.92, CH_2	1.98, m	39.93, CH_2	2.05, m	40.21, CH_2
	1.26, m		1.37, m		1.20, m		1.26, m		1.29, m	
23	4.90, d (7.8)	85.10, CH	5.37, d (8.7)	82.70, CH	4.82, d (8.5)	83.71, CH	4.83, d (8.0)	83.69, CH	4.91, d (8.5)	85.25, CH
24		163.32, C		164.42, C		161.08, C		161.06, C		163.58, C
25		123.44, C		122.92, C		123.13, C		123.11, C		123.48, C
26		176.87, C		174.99, C		174.87, C		174.84, C		177.15, C
27	1.79, s	8.39, CH_3	1.91, s	9.18, CH_3	1.77, s	8.17, CH_3	1.76, s	8.79, CH_3	2.01, s	12.35, CH_3
28	0.92, s	28.39, CH_3	1.02, s	28.34, CH_3	1.07, s	28.16, CH_3	1.02, s	28.37, CH_3	0.99, s	28.55, CH_3
29	0.96,s	22.30, CH_3	0.90,s	22.30, CH_3	0.89, s	22.33, CH_3	1.00,s	17.22, CH_3	1.29,s	18.78, CH_3
30	1.15, s	25.28, CH_3	1.41, s	25.53, CH_3	1.37, s	25.41, C	1.48, s	25.44, CH_3	1.11, s	24.97, CH_3
31	2.02, s	12.38, CH_3	4.84, d (15.0)	57.24, CH_2	1.84, s	12.26, CH_3	1.84, s	12.29, CH_3	1.77, s	8.30, CH_3
			4.67, d (15.0)							
1′		172.33, C		171.66, C		168.22, C		171.87, C		172.70, C
2′	2.71, d (15.2)	46.43, CH_2	3.08, d (14.8)	46.71, CH_2	3.69, s	43.42, CH_2	3.15, d (15.5)	47.45, CH_2	2.71, d (14.6)	46.79, CH_2
	2.66, d (15.2)		3.02, d (14.8)				3.08, d (15.5)		2.64, d (14.6)	
3′		70.69, C		70.20, C		170.62, C		70.51, C		70.80, C
4′	2.73, s	45.99, CH_2	2.99, d (14.5)	46.34, CH_2			3.10, s	47.45, CH_2	2.63, d (14.5)	46.42, CH_2
			2.97, d (14.5)						2.62, d (14.5)	
5′		175.10, C		172.28, C				174.88, C		172.71, C
6′	1.40, s	27.94, CH_3	1.67, s	28.78, CH_3			1.76, s	28.45, CH_3	1.37, s	27.74, CH_3
5′-OMe			3.64, s	51.67, CH_3						

[a] Recorded at 400 MHz for ^{1}H and 100 MHz f^{a13}C in Methanol-d_4.
[b] Recorded at 600 MHz for ^{1}H and 150 MHz f^{b13}C in Pyridine-d_5.
[c] Recorded at 600 MHz for ^{1}H and 150 MHz for^{13}C in Methanol-d_4; Chemical shifts and coupling constants (in parentheses) are given in ppm and Hz, respectively.

filtered to collect the H_2O-insoluble fraction (102 g). The filtrate was extracted with EtOAc (4 × 1 L) to give the EtOAc-soluble fraction (37 g). The H_2O-insoluble fraction and the EtOAc-soluble fraction were combined and separated on a silica gel column eluted with petroleum ether and EtOAc (40:1, 20:1, 10:1, 1:1, 0:1, v/v, each 1 L) to give fractions FPA1 - 13. FPA9 (20g) was separated over a silica gel column eluted with $CHCl_3$ - MeOH (120:1, 80:1, 40:1, 20:1, 10:1, 5:1, v/v, each 2 L) to give fractions FPA9A - D. FPA9A (610 mg) was separated by preparative HPLC with MeCN - H_2O (86:14, v/v, 4 mL/min) to give compounds **11** (13 mg, t_R 16.78 min), **6** (35 mg, t_R 24.55 min), and **9** (135 mg, t_R 36.31 min). FPA9 B (139 mg) was separated by semipreparative HPLC with MeCN - H_2O (98:2, v/v, 4 mL/min) to afford compounds **13** (48 mg, t_R 9.16 min), and **10** (18 mg, t_R17.76 min). FPA10 (31.6 g) was separated over a silica gel column eluted with petroleum ether and acetone (10:1, 5:1, 1:1, v/v, each 1 L) to give FPA10A - J. FPA10F (8.9 g) was separated by preparative HPLC with MeOH - H_2O (92:8, v/v, 10 mL/min) to give eight subfractions. Fraction FPA10F2 (297 mg) was separated by semipreparative HPLC with MeOH - H_2O (77.5:22.5, v/v, 4 mL/min) to afford compound **12** (106 mg, t_R 25.62 min). FPA10F4 (338 mg) was separated by semipreparative HPLC with MeOH - H_2O (83:17, v/v, 4 mL/min) to afford **14** (128 mg, t_R 28.76 min). Compounds **20** (3 mg, t_R 20.03 min) and **7** (9 mg, t_R 29.13 min) were isolated from fraction FPA10F5H (289 mg) by preparative HPLC with MeCN - H_2O (78:22, v/v, 4 mL/min). Compounds **16** (2.6 mg, t_R15.61 min) and **2** (98 mg, t_R16.75 min) were isolated from FPA10I (489 mg) by preparative HPLC with MeCN - H_2O (88:12, v/v, 4 mL/min). FPA11 (35 g) was separated over a reverse C_{18}-modified silica gel column with MeOH - H_2O (80:20, v/v, 10 mL/min) to give fractions FPA11A - F. FPA11C (1.26 g) was separated over a silica gel column eluted with

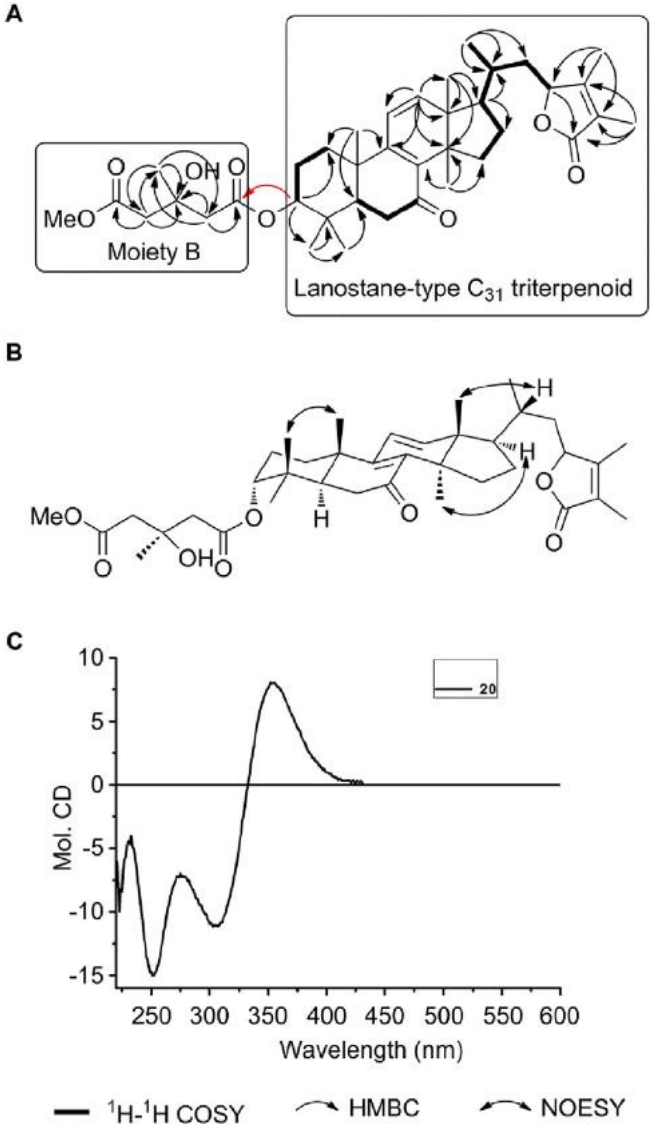

Fig. 5. ^{1}H-^{1}H COSY and key HMBC correlations (**A**), selected NOESY correlations (**B**), and ECD spectrum (**C**) of compound **20**.

$CHCl_3$ - MeOH (120:1, 80:1, 40:1, 20:1, 10:1, 5:1, v/v, each 1 L) to give fractions FPA11C1 - FPA11C5. Compounds **19** (18 mg, t_R 14.71 min) and **1** (108 mg, t_R 23.36 min) were isolated from FPA11C3 (439 mg) by semipreparative HPLC with MeCN - H_2O (63:37, v/v, with 0.1% formic acid, 4 mL/min). FPA11D (18.3 g) was separated over a silica gel column eluted with $CHCl_3$ - MeOH (100: 1, 50:1, 20:1, 10:1, 5:1, 1:1, v/v, each 3 L) to give fractions FPA11D1 - 6. Compounds **18** (13 mg, t_R 18.76 min) and **17** (10 mg, t_R 26.06 min), compound **15** (90 mg, t_R 26.56 min) and **8** (290 mg, t_R 29.56 min) were isolated from FPA11D2 (839 mg) by semipreparative HPLC with MeCN - H_2O (81:19, v/v, with 0.1% formic acid, 4 mL/min). Compounds **4** (17 mg, t_R 21.16 min), **3** (19 mg, t_R 26.03 min), and **5** (15 mg, t_R 27.69 min) were isolated from FPA11D3 (232 mg) by semipreparative HPLC with MeCN - H_2O (77:23, v/v, with 0.1% formic acid, 4 mL/min).

Palustrisoic acid A (**1**): white powder; $[\alpha]_D^{20}$ + 44 (*c* 0.6, MeOH); UV (MeOH) λ_{max} (logε) 213 (4.02) nm; IR (KBr) ν_{max} 3441, 3232, 2941, 1732, 1705, 1375, 1217, 1085, 998, and 565 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 669.3964 (calcd for $C_{37}H_{58}O_9$+Na^+, 669.3973), error 1.4 ppm.

Palustrisoic acid B (**2**): white powder; $[\alpha]_D^{20}$ + 68.6 (*c* 0.6, MeOH); UV (MeOH) λ_{max} (log ε) 202 (4.08) nm; IR (KBr) ν_{max} 3461, 2943, 1712, 1645, 1457, 1375, 1222, 988, and 595 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 653.4038 (calcd for $C_{37}H_{58}O_8$+Na^+, 653.4024), error −2.1 ppm.

Palustrisoic acid C (**3**): white powder; $[\alpha]_D^{20}$ + 54.5 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 204 (3.98) nm; IR (KBr) ν_{max} 3463, 2941, 1712, 1643, 1453, 1374, 1221, 9868, and 596 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 653.4043 (calcd for $C_{37}H_{58}O_8$+Na^+, 653.4024), error −2.9 ppm.

Palustrisoic acid D (**4**): white powder; $[\alpha]_D^{20}$ + 27.8 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 202 (4.02) nm; IR (KBr) ν_{max} 3432, 2941, 2882, 1715, 1700,1577, 1457, 1397, 1213, 1010, and 519 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 653.4007 (calcd for $C_{37}H_{58}O_8$+Na^+, 653.4024), error 2.6 ppm.

Palustrisoic acid E (**5**): white powder; $[\alpha]_D^{20}$ + 15 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 203 (4.18) nm; IR (KBr) ν_{max} 3430, 2941, 2880, 1715, 1700,1577, 1393, 1203, 1018, and 529 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 653.4024 (calcd for $C_{37}H_{58}O_8$+Na^+, 653.4024), error −0.1 ppm.

Palustrisoic acid F (**6**): white powder; $[\alpha]_D^{20}$ + 15.5 (*c* 0.2, MeOH); UV (MeOH) λ_{max} (log ε) 202 (4.33) nm; IR (KBr) ν_{max} 3747, 3445, 2949, 2875, 1716, 1437,1378, 1205, 1175, 898, and 584 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 2; HRESIMS *m/z* 665.4042 (calcd for $C_{38}H_{58}O_8$+Na^+, 665.4024), error −2.7 ppm.

Palustrisoic acid G (**7**): white powder; $[\alpha]_D^{20}$+65 (c 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 209 (4.10) nm; IR (KBr) ν_{max} 3431, 2936, 1722, 1637, 1379, 1204 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 3; HRESIMS *m/z* 509.3614 (calcd for $C_{31}H_{50}O_4$+Na^+, 509.3601), error 2.5 ppm.

Palustrisoic acid H (**11**): white powder; $[\alpha]_D^{20}$ - 7.3 (*c* 0.2, MeOH); UV (MeOH) λ_{max} (log ε) 203 (4.15) nm; IR (KBr) ν_{max} 3432, 2955, 1733, 1715, 1623,1379, 1199, 1178, 1014, and 571 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 667.4209 (calcd for $C_{38}H_{60}O_8$+Na^+, 667.4180), error −4.3 ppm.

Palustrisolide A (**14**): white powder; $[\alpha]_D^{20}$ + 8.2 (*c* 0.5, MeOH); UV (MeOH) λ_{max} (log ε) 210 (3.98) nm; ECD (MeOH) λ_{max} (Mol. CD) 223 (+5.74); IR (KBr) ν_{max} 3451, 3367, 2943, 1735, 1690, 1440, 1383, 1201, 1014, and 766 cm^{-1}; ^{1}H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 665.4038 (calcd for $C_{38}H_{58}O_8$+Na^+, 665.4024), error −2.1 ppm.

Palustrisolide B (**15**): white powder; $[\alpha]_D^{20}$ + 4.7 (*c* 0.6, MeOH);

Table 5
Cytotoxicity of the Isolated Lanostane-type C_{31} Triterpoid Derivatives.[a]

Compound	HCT116	A549	Hep[a]2	Compound	HCT116	A549	HepG2
cis-platin	18.7 ± 0.92	6.5 ± 0.43	1.9 ± 0.23	**12**	8.4 ± 0.49	12.1 ± 0.88	12.2 ± 0.80
6	86.7 ± 7.49	>100	>100	**13**	96.2 ± 5.31	>100	>100
8	92.7 ± 8.12	>100	>100	**14**	25.1 ± 1.18	46.3 ± 7.42	41.9 ± 3.67
9	86.2 ± 6.18	>100	>100	**15**	80.7 ± 7.37	>100	>100
10	96.6 ± 8.49	93.4 ± 7.30	88.9 ± 2.57	**16**	38.5 ± 2.06	44.6 ± 4.73	49.4 ± 3.25
11	>100	45.0 ± 6.81	>100	**20**	24.7 ± 1.50	29.6 ± 3.66	25.3 ± 3.11

[a] The IC_{50} values were stated in μM. The IC_5^a values of the other compounds for the tested cell lines are above 100 μM.

UV (MeOH) λ_{max} (log ε) 211 (4.35) nm; ECD (MeOH) λ_{max} (Mol. CD) 224 (+7.31); IR (KBr) ν_{max} 3421, 2941, 1733, 1716, 1600, 1378, 1115, 1053, 900, and 606 cm^{-1}; ^1H and ^{13}C NMR data, see Table 4; HRESIMS *m/z* 651.3883 (calcd for $C_{37}H_{56}O_8$+Na^+, 651.3867), error −2.4 ppm.

Palustrisolide C (**16**): white powder; $[\alpha]_D^{20}$ + 16.7 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 208 (4.12) nm; ECD (MeOH) λ_{max} (Mol. CD) 222 (+14.44); IR (KBr) ν_{max} 3441, 2941, 1733, 1451, 1378, 1202, and 1115 cm^{-1}; ^1H and ^{13}C NMR data, see Table 4; HRESIMS *m/z* 681.3989 (calcd for $C_{38}H_{58}O_9$+Na^+, 681.3973), error −2.4 ppm.

Palustrisolide D (**17**): white powder; $[\alpha]_D^{20}$ + 14.2 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 210 (4.0) nm; ECD (MeOH) λ_{max} (Mol. CD) 222 (+11.44); IR (KBr) ν_{max} 3423, 2945, 1733, 1610, 1380, 1165, 1053, and 606 cm^{-1}; ^1H and ^{13}C NMR data, see Table 4; HRESIMS *m/z* 593.3439 (calcd for $C_{34}H_{50}O_7$+Na^+, 593.3449), error 1.7 ppm.

Palustrisolide E (**18**): white powder; $[\alpha]_D^{20}$ + 9.1 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 210 (4.10) nm; ECD (MeOH) λ_{max} (Mol. CD) 226 (+7.93); IR (KBr) ν_{max} 3423, 2943, 1733, 1715, 1602, 1379, 1115, 1051, 901, and 607 cm^{-1}; ^1H and ^{13}C NMR data, see Table 4; HRESIMS *m/z* 651.3867 (calcd for $C_{37}H_{56}O_8$+Na^+, 651.3867), error 0.1 ppm.

Palustrisolide F (**19**): white powder; $[\alpha]_D^{20}$ + 21.3 (*c* 0.2, MeOH); UV (MeOH) λ_{max} (log ε) 213 (4.27) and 249 (3.91) nm; ECD (MeOH) λ_{max} (Mol. CD) 221 (+6.98); IR (KBr) ν_{max} 3441, 2936, 1732, 1716, 1636, 1379, 1214, 1091, and 599 cm^{-1}; ^1H and ^{13}C NMR data, see Table 4; HRESIMS *m/z* 667.3824 (calcd for $C_{37}H_{56}O_9$+Na^+, 667.3817), error −1.1 ppm.

Palustrisolide G (**20**): white powder; $[\alpha]_D^{20}$ + 30 (*c* 0.1, MeOH); UV (MeOH) λ_{max} (log ε) 212 (4.38) and 317 (3.91) nm; ECD (MeOH) λ_{max} (Mol. CD) 223 (−9.96), 252 (−15.20), 303 (−11.14), 352 (+8.03); IR (KBr) ν_{max} 3446, 2962, 2933,1733, 1716, 1657, 1369, 1149, and 571 cm^{-1}; ^1H and ^{13}C NMR data, see Table 1; HRESIMS *m/z* 661.3705 (calcd for $C_{38}H_{54}O_8$+Na^+, 661.3711), error 0.8 ppm.

3.4. X-ray crystallography

The white crystals of compounds **1** and **11** were obtained from MeOH-H_2O (30: 1, vol/vol) and MeOH, respectively. The X-ray diffraction data for **1** and **11** were collected at 293.15 K on a Bruker Xcalibur E Quest with an IμSmicrofocus source using Mo Kα radiation (λ = 0.71073 Å).

Crystallographic data for palustrisoic acid A (**1**): $C_{37}H_{58}O_9$; M = 646.83; crystal size 0.4 × 0.08 × 0.05 mm^3; monoclinic; *a* = 13.6922 (6) Å, *b* = 6.7453 (3) Å, *c* = 19.1422 (8) Å, β = 93.246 (4)°, V = 1765.10 (13) $Å^3$; space group $P2_1$, *Z* = 2; ρ_{calc} = 1.217 g/cm^3; μ = 0.085 mm^{-1}; 8603 reflections measured, 5839 unique (R_{int} = 0.0217), which were used for all calculations; the final refinement produced R_1 = 0.0597 (>2σ(I)) and $_wR_2$ = 0.1503 (all data) and Flack parameter = 0.3 (7).

Crystallographic data for palustrisoic acid H (**11**): $C_{38}H_{60}O_8$; M = 644.86; crystal size 0.4 × 0.06 × 0.04 mm^3; monoclinic, *a* = 7.5304 (4) Å, *b* = 15.3538 (7) Å, *c* = 31.8719 (13) Å, β = 90°, V = 3685.0 (3) $Å^3$; space group $P2_1$, *Z* = 4; ρ_{calc} = 1.162 g/cm^3; μ = 0.080 mm^{-1}; 22176 reflections measured, 7527 unique (R_{int} = 0.0515), which were used for all calculations; the final refinement produced R_1 = 0.0646 (>2σ(I)) and $_wR_2$ = 0.1353 (all data) and Flack parameter = 0.2 (7).

Crystallographic data for palustrisoic acids A (**1**, CCDC, 1819183) and H (**11**, CCDC, 1819185) have been deposited to the Cambridge Crystallographic Data Centre. These data can be obtained free of charge via http://www.ccdc.cam.ac.uk/conts/retrieving.html or from the CCDC, 12 Union Road, Cambridge CB2 1EZ, UK; Tel: (+44) 1223-336-408; Fax: (+44) 1223-336-033; or Email: deposit@ccdc.cam.ac.uk.

3.5. Alkaline hydrolysis of 2 and 3

The alkaline hydrolysis was performed on the basis of the reported procedure (Kim et al., 2013) with minor modification. Compounds **2** (23 mg) and **3** (8 mg) were hydrolyzed by 1 N NaOH solution at 80 °C for 2 h. The reaction mixture was adjusted to pH 6.0. Some solid was precipitated from the mixture. The solid was collected and identified. The triterpenoid acid of **2** is **7**, which was supported by HPLC analysis and NMR spectroscopic data comparison. Compound **3a** is the triterpenoid acid from **3**. Its structure was identified by NMR spectroscopic data (Fig. S1 in Supplementary Data and Table 3).

3.6. Cytotoxicity assay

The cytotoxicity against the human colorectal carcinoma cell line HCT116, the human lung carcinoma cell line A549, and the human hepatocellular carcinoma cell line HepG2 cell lines was assayed by using the MTT methods as previously reported (Yu et al., 2014). In brief, the tumor cells in logarithmic growth phase were incubated in 96-well microplates for 24 h at 37 °C in 5% CO_2 incubator. The isolated compounds and cis-platin were dissolved in DMSO with different concentrations. The samples were added to the cell culture and the resulting mixture was incubated for another 44 h. The buffer of the cell culture was replaced by 100 μL of fresh PBS and 10 μL of MTT (5 mg/mL) was added to each well and incubated for another 4 h. And then, 100 μL of 10% SDS was added to the cell culture. The absorbance of each well was determined by a microplate reader (Tecan, Research Triangle Park, NC, USA) at 570 nm. All experiments were performed in triplicate.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.phytochem.2018.04.012.

References

Allen, F.H., Kutney, J.P., Trotter, J., Westcott, N.D., 1971. The structures and absolute stereochemistry of cyclograndisolide and epicyclograndisolide, novel triterpene lactones from *Abies grandis*. Tetrahedron Lett. 12, 283–286.
Chiba, T., Sakurada, T., Watanabe, R., Yamaguchi, K., Kimura, Y., Kioka, N., Kawagishi, H., Matsuo, M., Ueda, K., 2014. Fomiroid A, a novel compound from the mushroom *Fomitopsis nigra*, inhibits NPC1L1-mediated cholesterol uptake via a mode of action distinct from that of ezetimibe. PLoS ONE 9, e116162.
De Silva, D.D., Rapior, S., Sudarman, E., Stadler, M., Xu, J., Alias, A., Hyde, K.D., 2013. Bioactive metabolites from macrofungi: ethnopharmacology, biological activities and chemistry. Fungal Divers. 62, 1–40.
He, J., Feng, X., Ji, T., Lu, Y., Zhao, B., 2006. Secondary metabolites from fungus *Fomes cajanderi*. Nat. Prod. Res. 20, 598–605.
Hill, R.A., Connolly, J.D., 2017. Triterpenoids. Nat. Prod. Rep. 34, 90–122.
Hwang, C.H., Jaki, B.U., Klein, L.L., Lankin, D.C., McAlpine, J.B., Napolitano, J.G., Fryling, N.A., Franzblau, S.G., Cho, S.H., Stamets, P.E., Wang, Y.H., Pauli, G.F., 2013. Chlorinated coumarins from the polypore mushroom *Fomitopsis officinalis* and their activity against *Mycobacterium tuberculosis*. J. Nat. Prod. 76, 1916–1922.
Jennings, J.P., Klyne, W., Scopes, P.M., 1965. Optical rotatory dispersion. part XXIV lactones. J. Chem. Soc. 1965, 7211–7229.
Jiang, Z., Tanaka, T., Hirata, H., Fukuoka, R., Kouno, I., 1977. The first euphane-type triterpene tridesmosides and bisdesmoside from *Rhoiptelea chiliantha*. Tetrahedron 53, 16999–17008.
Kamo, T., Asanoma, M., Shibata, H., Hirota, M., 2003. Anti-inflammatory lanostane-type triterpene acids from *Piptoporus betulinus*. J. Nat. Prod. 66, 1104–1106.
Keller, A.C., Maillard, M.P., Hostettmann, K., 1996. Antimicrobial steroids from the fungus *Fomitopsis pinicola*. Phytochemistry 41, 1041–1046.

Kim, K.H., Moon, E., Choi, S.U., Kim, S.Y., Lee, K.R., 2013. Lanostane triterpenoids from the mushroom *Naematoloma fasciculare*. J. Nat. Prod. 76, 845–851.
King, T.J., Smith, R.W., Begley, M.J., Goad, L.J., 1984. The stereochemistry of polyporenic acid a methyl ester. Tetrahedron Lett. 25, 3489–3492.
Konuma, R., Umezawa, K., Mizukoshi, A., Kawarada, K., Yoshida, M., 2015. Analysis of microbial volatile organic compounds produced by wood-decay fungi. Biotechnol. Lett. 37, 1845–1852.
Lai, K., Lu, M., Du, Y., El-Shazly, M., Wu, T., Hsu, Y., Henz, A., Yang, J., Backlund, A., Chang, F., Wu, Y., 2016. Cytotoxic lanostanoids from *Poria cocos*. J. Nat. Prod. 79, 2805–2813.
Li, H., Sun, N., Kashiwada, Y., Sun, L., Snider, J.V., Cosentino, L.M., Lee, K., 1993. Anti-AIDS agents, 9. Suberosol, a new C31 lanostane-type triterpene and anti-HIV principle from *Polyalthia suberosa*. J. Nat. Prod. 56, 1130–1133.
Li, L., Zhong, J., Tohtaton, Z., Ren, Q., Han, L., Hang, X., Yuan, T., 2016. Officimalonic acids A-H, lanostane triterpenes from the fruiting bodies of *Fomes officinalis*. Phytochemistry 130, 193–200.
Luo, G., Ye, Q., Du, B., Wang, F., Zhang, G., Luo, Y., 2016. Iridoid glycosides and diterpenoids from *Caryopteris glutinosa*. J. Nat. Prod. 79, 886–893.
Nes, W.D., 2011. Biosynthesis of cholesterol and other sterols. Chem. Rev. 111, 6423–6451.
Pérez-Gil, J., Rodríguez-Concepción, M., 2013. Metabolic plasticity for isoprenoid biosynthesis in bacteria. Biochem. J. 452, 19–25.
Pleszczyńska, M., Lemieszek, M.K., Siwulski, M., Wiater, A., Rzeski, W., Szczodrak, J., 2017. *Fomitopsis betulina* (formerly *Piptoporus betulinus*): the Iceman's polypore fungus with modern biotechnological potential. World J. Microb. Biotechnol. 33, 83.
Quang, D.N., Arakawa, Y., Hashimoto, T., Asakawa, Y., 2005. Lanostane triterpenoids from the inedible mushroom *Fomitopsis spraguei*. Phytochemistry 66, 1656–1661.
Rösecke, J., König, W.A., 2000. Constituents of the fungi *Daedalea quercina* and *Daedaleopsis confragosa* var. *tricolor*. Phytochemistry 54, 757–762.
Rösecke, J., König, W.A., 1999. Steroids from the fungus *Fomitopsis pinicola*. Phytochemistry 52, 1621–1627.
Sakurai, N., Goto, T., Nagai, M., Inoue, T., Xiao, P., 1990. Studies on the constituents of *Beesia calthaefolia* and *Souliea vaginata*. III. Beesioside IV, a cyclolampstanol xyloside from the rhizomes of *B. calthaefolia* and *S. vaginata*. Heterocyles 30, 897–904.
Sakurai, N., Nagai, M., Nagase, H., Kawai, K., Inoue, T., Xiao, P., 1986. Studies on the constituents of *Beesia calthaefolia* and *Souliea vaginata*. II. Beesioside II, a cyclolanostanol xyloside from rhizomes of *Beesia calthaefolia*. Chem. Pharm. Bull. 34, 582–589.
Schwab, W., Wüst, M., 2015. Understanding the constitutive and induced biosynthesis of mono- and sesquiterpenes in grapes (*Vitis vinifera*) - a key to unlocking the biochemical secrets of unique grape aroma profiles. J. Agric. Food Chem. 63, 10591–10603.
Shi, Z.T., Bao, H.Y., Feng, S., 2017. Antitumor activity and structure-activity relationship of seven lanostane-type triterpenes from *Fomitopsis pinicola* and *F. officinalis*. China J. Chin. Mater. Med. 42, 915–922.
Shibata, S., Natori, S., Fujuta, K., Kitagawa, I., Watanabe, K., 1958. Metabolic products of fungi. XV. Pachymic acid, a constituent of "Bukuryo" (Fu Ling茯苓), a sclerotium of *Poria cocos* (SCHW.) WOLF. Chem. Pharm. Bull. 6, 608–611.
Shingu, T., Tai, T., Akahori, A., 1992. A lanostane triterpenoid from *Poria cocos*. Phytochemistry 31, 2548–2549.
Tanaka, R., Wada, S., Aoki, H., Matsunaga, S., Yamori, T., 2004. Spiromarienonols A and B: two new 7(8→9)abeo-lanostane-type triterpene lactones from the stem bark of *Abies mariesii*. Helv. Chim. Acta 87, 240–249.
Tohtahon, Z., Xue, J., Han, J., Liu, Y., Hua, H., Yuan, T., 2017. Cytotoxic lanostane triterpenoids from the fruiting bodies of *Piptoporus betulinus*. Phytochemistry 143, 98–103.
Wang, L., Wang, N., Yao, X., Miyata, S., Kitanaka, S., 2003. Euphane and tirucallane triterpenes from the roots of *Euphorbia kansui* and their in vitro effects on the cell division of Xenopus. J. Nat. Prod. 66, 630–633.
Wangun, H.V.K., Berg, A., Hertel, W., Nkengfack, A.E., Hertweck, C., 2004. Anti-inflammatory and anti-hyaluronate lyase activities of lanostanoids from *Piptoporus betulinus*. J. Antibiot. 57, 755–758.
Wu, X., Yang, J., Yan, M., 2009. Four new triterpenes from fungus of *Fomes officinalis*. Chem. Pharm. Bull. 57, 195–197.
Yoshikawa, K., Inoue, M., Matsumoto, Y., Sakakibara, C., Miyataka, H., Matsumoto, H., Arihara, S., 2005. Lanostane triterpenoids and triterpene glycosides from the fruit body of *Fomitopsis pinicola* and their inhibitory activity against COX-1 and COX-2. J. Nat. Prod. 68, 69–73.
Yu, M., Luo, X., Wei, W., Xu, X., Jiang, N., 2014. CN Patent 103877126.